

Exact endomorphisms and Hodge classes for a split abelian-cover family

Revised Phase I: proved scope and general repair boundary

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Abstract

For every odd prime p , we give a proof of the generic isogeny decomposition and endomorphism algebra of the family $u^p = x/(x - t)$, $v^p = x - 1$. Two local braid eigenvalue ratios determine its independent rank-two monodromy blocks; explicit involutions supply all multiplicity maps. Consequently the rational Hodge ring of every power of a very general Jacobian is generated by divisors. The proof uses explicitly matched cyclic-cover results of Menet–Nguyen and classical Hodge and invariant theory. It does not use the missing legacy Fox–Gassner certificate or a Phase II fusion theorem. We also correct the Galois scope of the moving part and the orientation of quotient-graph maps. The former blanket general finite-abelian block and cross-Hom claims are not asserted here. This is a mathematical manuscript, not a claim of Lean verification or external peer review.

1 The restricted theorem and its inputs

Let $B = \mathbb{P}_{\mathbb{C}}^1 \setminus \{0, 1, \infty\}$, let $p \geq 3$ be prime, and let $X_{p,t}$ denote the smooth projective normalization of

$$u^p = \frac{x}{x - t}, \quad v^p = x - 1, \quad t \in B. \quad (1)$$

Write $J_t = JX_{p,t}$, $K = \mathbb{Q}(\zeta_p)$, $F = K^+$, and $d = [F : \mathbb{Q}] = (p - 1)/2$. A statement about a very general fibre excludes a countable union of proper closed subsets of B ; it does not describe all special fibres.

Theorem 1.1. *For a very general t , there are pairwise non-isogenous absolutely simple complex abelian varieties B_ω , indexed by $\Omega_p = \mathbb{F}_p^\times / (r \sim r^{-1})$, of dimension d , such that*

$$J_t \sim B_{\{1\}}^2 \times B_{\{-1\}}^2 \times \prod_{\substack{\omega \in \Omega_p \\ |\omega|=2}} B_\omega^4, \quad (2)$$

$$\text{End}^0(B_\omega) = F, \quad \text{End}^0(J_t) = M_2(F)^2 \times M_4(F)^{(p-3)/2}, \quad (3)$$

$$\text{Hg}(J_t)^{\text{der}} = (\text{Res}_{F/\mathbb{Q}} \text{SL}_2)^{(p+1)/2}, \quad (4)$$

$$\text{rank NS}(J_t) = \frac{(p-1)(5p-9)}{2}. \quad (5)$$

The genus is $(p-1)^2$. For every $q \geq 1$, all rational Hodge classes on J_t^q are generated by divisor classes and are algebraic.

The cyclic-cover input is [1], with its standing restriction $N \geq 3$, $n \geq 3$, finite labels $1 \leq k_i < N$, $\gcd(k_1, \dots, k_n, N) = 1$, and primitive $q_0 = \exp(-2\pi i k/N)$, $\gcd(k, N) = 1$. Theorem 2.1 gives eigenspace dimension and signature. Definition 1.1(a) and Theorem 5.1 give connected special-unitary monodromy when $1 < \sum_i \{k k_i/N\} < n - 1$. Theorem 2.5 and the reflection calculation immediately before Corollary 2.7 give eigenvalues $1, q_0^{k_i+k_j}$ in rank two; when the second is one, Corollary 2.7 gives unipotence. We do not infer nonidentity from unipotence. This input is a preprint theorem, not a theorem proved by the certificate below.

The other inputs are semisimplicity of geometric polarized variations, the inclusion of connected monodromy in the very general Hodge group [3, 4], the equivalence between rational weight-one Hodge maps and isogeny-category maps of complex abelian varieties, the Néron–Severi/Rosati correspondence [5], and the first fundamental theorem for the standard representation of SL_2 [6]. The algebraic-group and representation calculations needed here are given below.

2 Geometry and algebraic involutions

Let ρ_1, ρ_2 multiply u, v by ζ_p . The two defining Kummer functions are independent modulo p -th powers: their valuations at 0 and 1 are respectively $(1, 0)$ and $(0, 1)$. Thus the cover is connected of degree p^2 , with group $\mathbb{F}_p^2$ and inertia of order p at the four marked points. Riemann–Hurwitz gives

$$2g - 2 = p^2(-2 + 4(1 - 1/p)), \quad g = (p - 1)^2.$$

For $a, b \in \mathbb{F}_p$, the cyclic quotient is represented by $y = u^a v^b$; its four residues are $(a, b, -a, -b)$. If exactly one of a, b vanishes, it has two branch points and genus zero. If both are nonzero, choose $1 \leq a, b < p$. The finite labels at $0, 1, t$ can be taken to be

$$(a, b, p - a). \tag{6}$$

Multiplying y by $x - t$ accounts for the replacement of $-a$ by $p - a$. Infinity has residue $-b$.

Let $W_{a,b}$ denote the $q_0 = \zeta_p^{-1}$ eigenspace on this quotient, pulled to X . As a G -character it has values $\rho_1 \mapsto \zeta_p^{-a}, \rho_2 \mapsto \zeta_p^{-b}$; this fixes the root-sign convention once and for all. At every nonzero row k ,

$$\{ka/p\} + \{kb/p\} + \{k(p - a)/p\} = 1 + \{kb/p\} \in (1, 2). \tag{7}$$

Theorem 2.1 of [1] gives rank two and signature $(1, 1)$. These $W_{a,b}$, for $a, b \neq 0$, exhaust $H^1(X, \mathbb{C})$. In particular all their Galois conjugates are noncompact factors.

Pass to the finite étale base cover $s^p = t - 1$. On it define

$$\Phi(x, u, v) = \left(\frac{t(x - 1)}{x - t}, \frac{v}{s}, su \right), \tag{8}$$

$$\Psi(x, u, v) = \left(\frac{t - x}{1 - x}, -\frac{1}{su}, -\frac{s}{v} \right), \tag{9}$$

$$\Omega(x, u, v) = \left(\frac{t}{x}, -\frac{1}{v}, -\frac{1}{u} \right). \tag{10}$$

Put $f_1 = x/(x - t)$, $f_2 = x - 1$, and denote the three first coordinates by T_Φ, T_Ψ, T_Ω . The identities

$$\begin{aligned} f_1(T_\Phi) &= f_2/(t - 1), & f_2(T_\Phi) &= (t - 1)f_1, \\ f_1(T_\Psi)f_1 &= -1/(t - 1), & f_2(T_\Psi)f_2 &= 1 - t, \\ f_1(T_\Omega)f_2 &= -1, & f_2(T_\Omega)f_1 &= -1 \end{aligned}$$

prove preservation of the equations, using that p is odd. They induce isomorphisms of function fields and hence of the smooth projective curves. Direct substitution gives

$$\Phi^2 = \Psi^2 = \Omega^2 = 1, \quad \Phi\Psi = \Psi\Phi = \Omega.$$

Conjugation on the deck group gives, on character labels,

$$\Phi : (a, b) \mapsto (b, a), \quad \Psi : (a, b) \mapsto (-a, -b), \quad \Omega : (a, b) \mapsto (-b, -a). \quad (11)$$

Their graphs are algebraic correspondences over the base cover. All monodromy assertions below concern the identity component, which is unchanged by this finite base change.

3 Exact rank-two independence from two eigenvalue ratios

Lemma 3.1. *Each $W_{a,b}$ has connected algebraic monodromy SL_2 over $\mathbb{C}$, and connected projective real Zariski closure $\mathrm{PSL}_2(\mathbb{R})$.*

Proof. Apply the source input to $N = p, n = 3$ and (6). Every label is active, the gcd condition holds since $a \neq 0$, and (7) verifies the good condition with the source's negative exponential convention. Its signature is $(1, 1)$. The theorem is initially on the ordered configuration of three finite points (b_1, b_2, b_3) . Put

$$\alpha = b_2 - b_1, \quad \beta = b_1, \quad t = (b_3 - b_1)/\alpha, \quad x = \alpha z + \beta.$$

This identifies the configuration space with $\mathrm{Aff}_1 \times B$. On the entire abelian-cover family the equations become

$$u^p = \frac{x - b_1}{x - b_3} = \frac{z}{z - t}, \quad v^p = x - b_2 = \alpha(z - 1).$$

Thus u is unchanged and $v = \alpha^{1/p} v_{\mathrm{norm}}$. Monodromy on the affine-coordinate fibre is a deck transformation, acting as a finite-order scalar on each character eigenspace. Consequently the full joint representation into the product of the projective groups of the $W_{a,b}$ factors through B . The same braids α_{12}, α_{23} can therefore be tested before or after normalization, simultaneously for every pair of factors. In particular the connected projective images are preserved. The determinants of braid generators are p -th roots of unity by Corollary 2.7; the same holds after normalization. Thus the connected linear image is $\mathrm{SU}(1, 1)$, whose complexification is SL_2 . $\square$

Lemma 3.2 (Normalizer of a graph). *Let H be a connected centreless simple algebraic group in characteristic zero. For an automorphism θ of H , the normalizer in $H \times H$ of $\Gamma_\theta = \{(h, \theta(h)) : h \in H\}$ is Γ_θ . If two projective monodromy factors form a connected graph, their full projective representations obey that graph relation.*

Proof. If (a, b) normalizes the graph, then for every h , $b\theta(h)b^{-1} = \theta(aha^{-1})$. Therefore $\theta(a)^{-1}b$ centralizes H and is the identity. For the second assertion, the identity component of an algebraic closure is normal in the full closure; apply the first assertion to the two-coordinate closure. In particular one must not raise a finite-order braid image to a power and then infer a graph obstruction from the resulting identity. $\square$

Lemma 3.3. *Two split character factors can lie in one connected projective monodromy graph only if*

$$(a', b') \in \{(a, b), (-a, -b), (b, a), (-b, -a)\}. \quad (12)$$

Proof. Use the same ordered finite labels $0, 1, t$ for both factors. For the pure braids α_{12} and α_{23} , the source reflection formula gives unordered eigenvalue ratios

$$\{\zeta_p^{a+b}, \zeta_p^{-(a+b)}\}, \quad \{\zeta_p^{b-a}, \zeta_p^{-(b-a)}\}.$$

The displayed signs absorb both the fixed negative-root convention and the ordering of the two eigenvalues. This assertion also holds when a ratio is one; no nonidentity assertion is used. By Lemma 3.2 a connected graph forces equality of these projective conjugacy invariants. Algebraic automorphisms of PSL_2 over $\mathbb{C}$ are inner, so they preserve these invariants. For independent signs $\epsilon, \eta \in \{1, -1\}$, we obtain

$$a' + b' = \epsilon(a + b), \quad a' - b' = \eta(a - b) \quad \text{in } \mathbb{F}_p.$$

As 2 is invertible,

$$a' = \frac{\epsilon + \eta}{2}a + \frac{\epsilon - \eta}{2}b, \quad b' = \frac{\epsilon - \eta}{2}a + \frac{\epsilon + \eta}{2}b.$$

Equal signs give $(a', b') = \epsilon(a, b)$, and opposite signs give $(a', b') = \epsilon(b, a)$. This includes $a = b$ and $a = -b$. $\square$

Proposition 3.4. *The connected complex monodromy is a product of standard SL_2 factors, one for each orbit (12). Each such factor acts diagonally on the rank-two summands in that orbit and trivially on the other orbits. There are $(p^2 - 1)/4$ factors: $p - 1$ orbits of size two and $(p - 1)(p - 3)/4$ orbits of size four.*

Proof. Semisimplicity and the surjections in Lemma 3.1 imply the usual semisimple Goursat decomposition into graph blocks. Indeed each simple Lie ideal projects either to zero or isomorphically to a simple target; two different ideals cannot both project nontrivially to one target, since their images commute. Lemma 3.3 excludes graphs between distinct displayed orbits. Conversely (11) supplies isomorphisms of local systems throughout each orbit after the finite base change, forcing the same monodromy action there. Connected determinants are trivial; lifting a projective graph gives the diagonal SL_2 on the standard modules. The standard two-dimensional representation is faithful, so no additional quotient or connected scalar factor remains.

Only the lines $a = b$ and $a = -b$ have size-two orbits, giving $p - 1$ such orbits. All other orbits have size four. Equivalently Burnside gives $((p - 1)^2 + 2(p - 1))/4 = (p^2 - 1)/4$. $\square$

Remark 3.5. This proof uses neither an exact Fricke trace product nor a half-shift exclusion at even denominators, and does not depend on the missing Fox–Gassner source. It proves this split odd-prime case; it is not a replacement for those calculations for arbitrary disk words.

4 Algebraic matrix units and the rational commutant

Lemma 4.1 (Use inverse maps, not an oriented Rosati shortcut). *Suppose algebraic correspondences give isomorphisms $T_i : W_0 \rightarrow W_i$, with algebraic inverses $S_i : W_i \rightarrow W_0$. Include the character projectors in these maps, so all maps vanish off their stated source. Then $E_{ij} = T_i S_j$ satisfy $E_{ij} E_{k\ell} = \delta_{jk} E_{i\ell}$.*

Proof. The middle composite $S_j T_k$ is zero for $j \neq k$, and is 1_{W_0} for $j = k$. All compositions remain algebraic. For an automorphism of X , its inverse graph gives S_i . For a quotient $q : X \rightarrow Y$, the identity $q_* q^* = (\deg q) 1$ gives the required rational normalization. No square roots of Schur scalars are needed. $\square$

Remark 4.2. For the bilinear polarization, the cyclotomic-linear extension of Rosati satisfies $e_c^\dagger = e_{-c}$. The adjoint of an isolated arrow $W_c \rightarrow W_e$ has source W_{-e} , not in general W_e . Thus the old formula using $\Theta_i^\dagger \Theta_i$ on an oriented corner cannot justify these matrix units. Explicit inverse graphs do.

Rational deck-isotypic factors are indexed by $L_r = \mathbb{F}_p(1, r)$, $r \in \mathbb{F}_p^\times$. Denote their abelian varieties by A_r ; their dimension is $p-1$, since each has $p-1$ complex eigenspaces of rank two. The rational deck algebra on A_r is K . The algebraic involution Ψ preserves A_r , has square one, and conjugates K by complex conjugation. Hence there is a unital algebra map

$$K \rtimes \text{Gal}(K/F) \longrightarrow \text{End}^0(A_r).$$

The crossed product is $\text{End}_F(K) = M_2(F)$: multiplication by K and conjugation act faithfully on the two-dimensional F -space K , and both algebras have dimension four over F . Its simplicity makes the unital map injective. A primitive idempotent gives $A_r \sim B_r^2$, with $\dim B_r = d$. The map Φ identifies A_r with $A_{r^{-1}}$. For each nonfixed pair, its graph and inverse combine the two $M_2(F)$ actions into $M_4(F)$; possible twists of the identified copy of F do not change this algebra isomorphism. Thus there is an algebraic subalgebra

$$\mathcal{A} = M_2(F)^2 \times M_4(F)^{(p-3)/2} \subseteq \text{End}^0(J_t). \quad (13)$$

All rational idempotents and inverses here are in the isogeny category.

Proposition 4.3. *For the connected monodromy group M , $\text{End}_M(H^1(J_t, \mathbb{Q})) = \mathcal{A}$ (with the harmless opposite algebra convention for pullback on H^1).*

Proof. Over $\mathbb{C}$, Proposition 3.4 and Schur's lemma give exactly one matrix algebra $M_m(\mathbb{C})$ for each oriented orbit of size $m = 2$ or 4 , and no maps between different orbits. Lemma 4.1, applied to Φ, Ψ, Ω , supplies all of these matrix units algebraically. Equivalently the complex dimension of the commutant is

$$4(p-1) + 16 \frac{(p-1)(p-3)}{4} = 4(p-1)(p-2).$$

This equals the rational dimension of $\mathcal{A}$ in (13). The algebraic correspondences commute with connected monodromy after the finite base change. Equality after scalar extension, or equality of these dimensions and the inclusion, proves the rational assertion. No unsupported descent of a chosen single cyclotomic arrow is used. $\square$

For a very general fibre the connected monodromy group lies in the Hodge group, so

$$\mathcal{A} \subseteq \text{End}_{\text{HS}}(H^1) \subseteq \text{End}_M(H^1) = \mathcal{A}.$$

Weight-one full faithfulness gives (3). Its primitive idempotents give (2). Different ω have no Hom corner and are not isogenous; $\text{End}^0(B_\omega) = F$ has no nontrivial idempotent, so B_ω is simple, and hence absolutely simple over $\mathbb{C}$.

The centralizer of $\mathcal{A}$ in the rational symplectic group is

$$C_{\text{Sp}}(\mathcal{A}) = \prod_{\omega \in \Omega_p} \text{Res}_{F/\mathbb{Q}} \text{Sp}_2 = \prod_{\omega \in \Omega_p} \text{Res}_{F/\mathbb{Q}} \text{SL}_2.$$

To check its form, on one component write $H^1 = F^2 \otimes_F F^{m_\omega}$. Positivity of Rosati fixes the totally real centre F and makes its involution on the split matrix algebra $M_{m_\omega}(F)$ orthogonal: it is adjoint to a nondegenerate symmetric multiplicity form. The polarization is therefore the trace of the

product of that form with an alternating form on F^2 . The operators commuting with $M_{m_\omega}(F)$ therefore preserve this nondegenerate rank-two alternating form. Every such form has a symplectic basis over F , so its group is the split $\mathrm{Sp}_2 = \mathrm{SL}_2$. This centralizer is connected. By Proposition 3.4, M has the same complexification and dimension as this centralizer, hence equals it over $\mathbb{Q}$. Algebraicity of $\mathcal{A}$ now gives

$$M \subseteq \mathrm{Hg} \subseteq C_{\mathrm{Sp}}(\mathcal{A}) = M.$$

In particular (4) holds; this argument also shows that the Hodge group itself is semisimple in this restricted family.

On $M_m(F)$, a positive Rosati involution is adjoint with respect to a symmetric form over F . Its fixed subspace has dimension $d m(m+1)/2$ over $\mathbb{Q}$. The Néron–Severi/Rosati correspondence therefore gives

$$\mathrm{rank} \, \mathrm{NS}(J_t) = d \left(2 \cdot 3 + \frac{p-3}{2} \cdot 10 \right) = \frac{(p-1)(5p-9)}{2}.$$

5 All powers

Proof of the last assertion of Theorem 1.1. Fix $q \geq 1$. After scalar extension, $H^1(J_t^q)$ is a direct sum of standard two-dimensional modules for the independent SL_2 factors, with their multiplicities multiplied by q . The first fundamental theorem for SL_2 states that tensor invariants are spanned by products of alternating pair contractions. Passing to the exterior algebra by antisymmetrization shows that its invariant algebra is generated by invariant classes of degree two.

Every multiplicity map is algebraic by Proposition 4.3, including the matrix maps between the q copies. Together with a polarization these give the invariant bilinear pairings. Their alternating parts are algebraic divisor classes, equivalently the corresponding Poincaré classes on the products and their pullbacks. Thus the entire M -invariant exterior algebra is generated by divisors. Since $M \subseteq \mathrm{Hg}$, all rational Hodge classes are among these invariants. Conversely divisors are Hodge classes. Equality follows over $\mathbb{C}$ and then over $\mathbb{Q}$ by faithful scalar extension. This uses no central-torus or fusion theorem. $\square$

6 Correct moving parts for general abelian branch data

This section repairs a general reduction, without asserting a new complete general block theorem. Let a full family of connected finite abelian covers of $\mathbb{P}^1$ have rational cohomology V and connected monodromy M . Its deck algebra decomposes $V = \bigoplus_{\mathcal{O}} V_{\mathcal{O}}$ into rational character orbits. An orbit means a full cyclotomic Galois orbit, not just the pair $\{c, -c\}$. For a character with s nonzero branch residues $0 < \alpha_i < 1$, set $Q(c) = \sum_i \alpha_i \in \mathbb{Z}$. Its two Hodge multiplicities, in one of the two orientations, are $s - 1 - Q(c)$ and $Q(c) - 1$. Call the orbit positive if some character in it has both multiplicities positive.

Proposition 6.1 (Galois-saturated moving criterion). *Assume that every positive character in the family has full connected factorwise monodromy on its eigenspace, and hence no invariant vector. Then, after a finite étale base change,*

$$V^M = \bigoplus_{\mathcal{O} \text{ not positive}} V_{\mathcal{O}}, \quad V_{\mathrm{mov}} = (V^M)^{\perp} = \bigoplus_{\mathcal{O} \text{ positive}} V_{\mathcal{O}}.$$

All complex embeddings of a positive rational orbit must be retained, including its definite compact embeddings.

Proof. The M -fixed subspace is rational and stable under the deck field of each $V_{\mathcal{O}}$. At a positive embedding its scalar extension is zero by the stated projection premise. A vector space over that deck field has equal dimension at all embeddings, so this fixed subspace is zero on the whole rational orbit.

If no character in an orbit is positive, every real character pair has definite Hodge signature. The real deck centralizer preserving the polarization is then a product of compact unitary groups. The whole rational summand, unlike an isolated complex eigenspace, has a monodromy-stable integral lattice, obtained by intersection with the integral cohomology lattice. Its monodromy lies in a compact group and in a discrete integral matrix group, hence is finite. A common finite étale cover kills these finitely many finite images. The deck decomposition is orthogonal between rational orbits, proving the stated orthogonal-complement formula. $\square$

The projection premise has a precise source match for characters of effective order $N \geq 3$ on the full branch family. Put one active branch point at infinity. The $n = s - 1$ finite weights have sum $Q(c) - \alpha_{\infty}$. Positivity gives

$$1 < Q(c) - \alpha_{\infty} < s - 2 = n - 1.$$

Their integral labels are nonzero, the effective-order condition gives the gcd equal to one, and either primitive orientation satisfies the same strict inequalities. These are the hypotheses of [1, Definition 1.1(a), Theorem 5.1]. Forgetting inactive labels is a configuration-space fibration with connected fibre and surjects on fundamental groups. Normalization of the active labels changes only finite scalar monodromy, as above. Effective order two uses the separate hyperelliptic symplectic monodromy input of A'Campo [2], not Theorem 5.1. Thus the proposition is an exact general representation-theoretic repair; it does not by itself determine graph identifications between different rational orbits.

The old lemma discarding each definite complex eigenspace is false. For example $c = (1, 1, 1, 2)/5$ has signatures, over its four embeddings, $(2, 0), (1, 1), (1, 1), (0, 2)$. The second embedding has full noncompact monodromy. If the first had finite image, applying the cyclotomic field automorphism to a basis over $\mathbb{Q}(\zeta_5)$ would make the second image finite too. This contradicts full projection.

7 Correctly oriented general quotient arrows and boundaries

For four active labels, let P be a double transposition, and let cyclic words c, e have the same effective order N . Fix the same primitive root q_0 when defining the eigenspaces on $Y_c : y^N = f_c(x)$ and $Y_e : v^N = f_e(x)$. If $e = Pc$, the divisor identity

$$f_e(T_P(x))/f_c(x) = a(t)h(x)^N$$

gives, after adjoining a root of the unit $a(t)$, an algebraic deck-equivariant map

$$(x, y) \longmapsto (T_P(x), a(t)^{1/N} h(x)y).$$

It identifies W_c with W_e . The unit is a constant times integral powers of $t, t - 1$, from the displayed double-transposition Möbius functions, so the base change is étale over B .

If $e = -Pc$, the analogous map has second coordinate proportional to $h(x)/y$. It conjugates the cyclic generator to its inverse, so the q_0 eigenspace on Y_c maps to the q_0^{-1} eigenspace on Y_e , that is to W_{-e} , not automatically W_e . Hence geometric oriented arrows are indexed by the actual relations $e = Pc$. A relation only up to sign classifies possible projective equivalence, not a specified Hodge Hom corner.

For a genuine oriented arrow, pull-graph-push and the normalization $(\deg q_c)^{-1}q_{c*}$ give an isomorphism on W_c . The inverse graph and $(\deg q_e)^{-1}q_{e*}$ give its inverse. Lemma 4.1 therefore proves matrix-algebra saturation on each actually connected oriented graph component. Taking the Galois-stable algebra of these correspondences permits rational descent, but supplies no arrows between disconnected oriented components.

In particular, a missing internal standard-dual map can be transported across an existing graph to a missing cross-coordinate map. It is incorrect to exclude internal corners from a theorem and then assert that every off-diagonal M -intertwiner is a graph Hodge map. In (1), the explicit Ψ supplies precisely the needed orientation maps, which is why the restricted theorem is unaffected.

The following former claims are withdrawn as unconditional statements in this revision: the real monodromy product obtained by deleting all definite complex eigenspaces; the blanket equality of graph algebras with all cross-character End_M corners; and the complete finite-abelian Hodge-group or all-powers theorem. Reinstating a general result requires an exact block theorem on the corrected rational representation (including compact embeddings), correctly oriented algebraic generators, and a proof of the remaining determinant-tensor algebraicity. General support separation, high-rank reconstruction, and the even-denominator four-point fingerprint argument retain their own proof/source obligations; none is silently treated as an external theorem here. Phase II's global fusion and cycle-specialization obligations remain separate.

A Reproducible checks and provenance

The new `scripts/check_phase_i.py` is a standalone symbolic and finite regression script using SymPy. It checks the six rational identities, all involution and deck-conjugation relations, the two-eigenratio fibres, the source inequalities, and the orbit and dimension formulas for every odd prime up to a configurable bound (default 101). It includes small exact crossed-product and inverse-matrix-unit checks and records the new source hashes. These finite checks are not the proof for all primes, nor a formalization of monodromy or Hodge theory.

The original manuscripts and `legacy/certificates/` are unchanged provenance copies. In particular the absent source `phase1src/p_ary_mt_defect_certificate.py` has not been reconstructed or represented as recovered. The frozen 7,077,120-vector and Fox-Gassner claims are not outputs of the new script. The restricted proof above no longer needs them.

References

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